I.

1.

Reference solution: $y = C - \dfrac{2\pi^2 k}{mg}(x-a)^2$ （C is an arbitrary constant).

2.

Reference solution: $\dfrac{7}{12}R$.

3.

Reference solution: $30\ \mu\text{m}$.

4.

Reference solution:

$U_{\text{O}} = \dfrac{kT}{e}\ln\left(\dfrac{I_{\text{SC}}}{I_{\text{S}}}+1\right)$; e is electron charge magnitude, k Boltzmann's constant, and T is absolute temperature; I_{S} is the reverse saturation current of the p-n junction.

II.

Reference solution:

1. Scaling position and time as $\boldsymbol{r}' = \alpha \boldsymbol{r}$ 、 $t' = \beta t$ gives velocity and acceleration:

$$\boldsymbol{v}' = \lim_{\Delta t' \to 0} \frac{\Delta \boldsymbol{r}'}{\Delta t'} = \frac{\alpha}{\beta} \lim_{\Delta t \to 0} \frac{\Delta \boldsymbol{r}}{\Delta t} = \frac{\alpha}{\beta}\boldsymbol{v} \ , \tag{1}$$

$$\boldsymbol{a}' = \lim_{\Delta t' \to 0} \frac{\Delta \boldsymbol{v}'}{\Delta t'} = \frac{\alpha}{\beta^2} \lim_{\Delta t \to 0} \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{\alpha}{\beta^2}\boldsymbol{a} \ . \tag{2}$$

Both the original and scaled particle motion obey Newton's equations:

$$m\boldsymbol{a} = \boldsymbol{F}(\boldsymbol{r}) \ , \tag{3}$$

$$m\boldsymbol{a}' = \boldsymbol{F}(\boldsymbol{r}') \ . \tag{4}$$

Using (2) and $\boldsymbol{F}(\alpha \boldsymbol{r}) = \alpha^k \boldsymbol{F}(\boldsymbol{r})$, (4) becomes

$$m\boldsymbol{a} = \alpha^{k-1}\beta^2 \boldsymbol{F}(\boldsymbol{r}) \ . \tag{5}$$

Comparison with (3) gives

$$\beta = \alpha^{\frac{1-k}{2}} \ . \tag{6}$$

2. For gravity, $k = -2$. If the orbital size scales as

$$l' = \alpha l \tag{7}$$

its period scales as